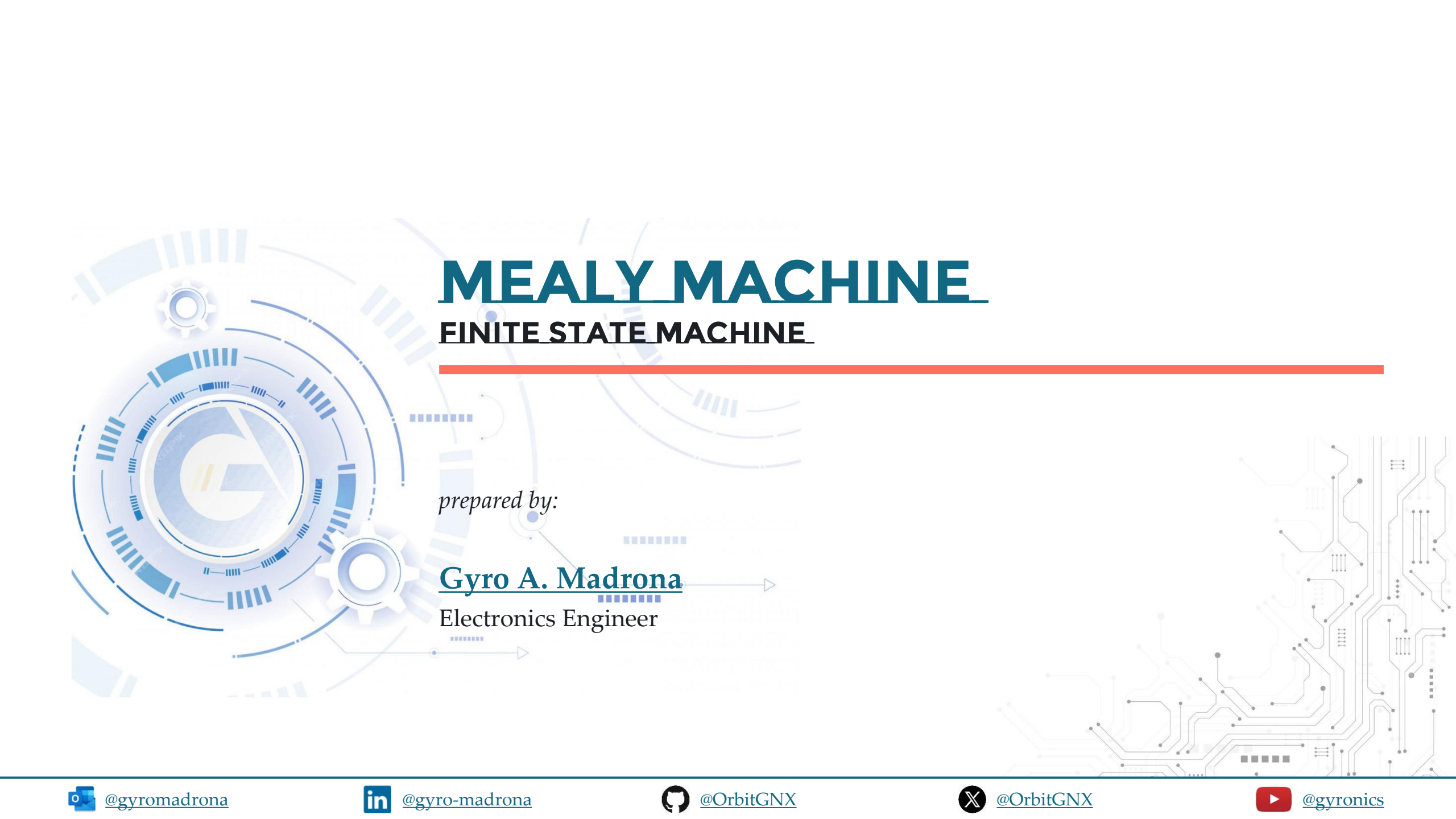




MEALY MACHINE

FINITE STATE MACHINE

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TOPIC OUTLINE

Mealy Machine

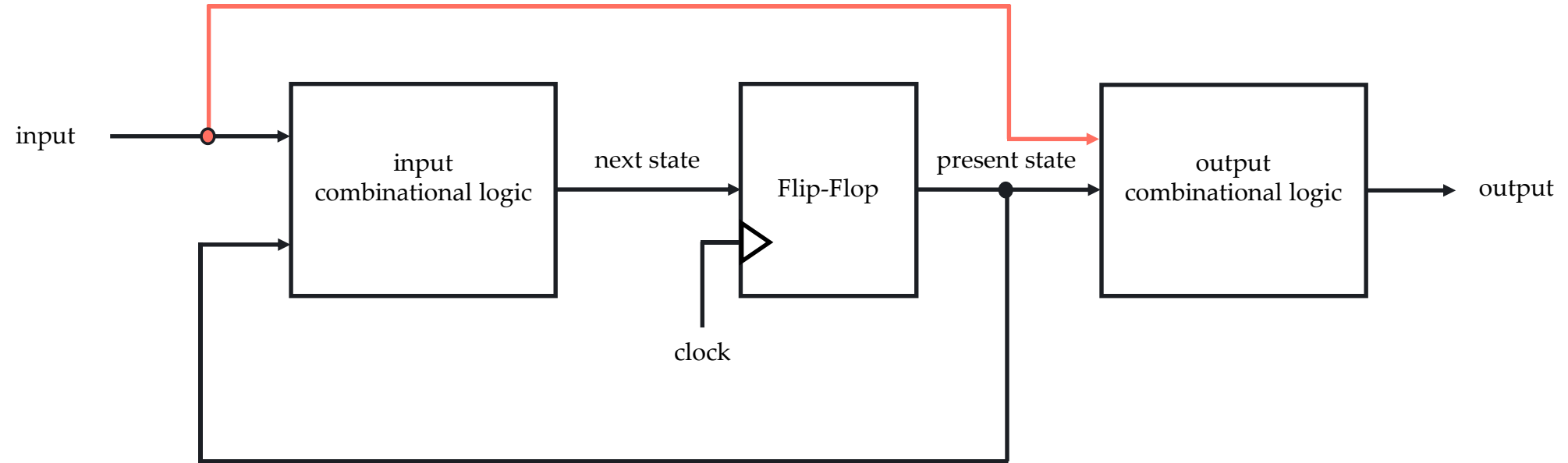
- Sequence 1011
- Sequence 0110



MEALY MACHINE



MEALY MACHINE



A Mealy machine is a type of finite-state machine in which the outputs depend on the present state and the input.



SEQUENCE 1011

This Mealy FSM detects the non-overlapping sequence 1011, assuming the input stream arrives LSB-first.

System Level Design

```
module seq_1011( );
```

```
endmodule
```



SEQUENCE 1011

State Diagram



SEQUENCE 0110

This Mealy FSM detects the overlapping sequence 0110, assuming the input stream arrives LSB-first.

System Level Design

```
module seq_0110( );
```

```
endmodule
```



SEQUENCE 0110

State Diagram



LABORATORY

